

Gradescope: To submit this homework assignment, use Gradescope accessed through Canvas. The assignment could be either typed using LaTeX or handwritten and scanned. Please allow some time in advance of the deadline to become familiar with Gradescope.

Challenge: For optional challenge problems, you will receive the specific points for each problem if your answer is correct, and you will not lose points for just ignoring them. **Note:** The total score for this assignment will not exceed 100.

Problem 1 (Linear Algebra Review, 30 points) Many of the ML algorithms we will study are based on concepts from linear algebra. This problem reviews some key concepts. For each expression below, evaluate the expression and either simplify or expand into a matrix as instructed.

- (a) (8 points) Given a matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ and a column vector $\mathbf{v} \in \mathbb{R}^n$, prove $(\mathbf{A}\mathbf{v})^T = \mathbf{v}^T \mathbf{A}^T$. Hint:

Write $\mathbf{A}$ in terms of row vectors, i.e., $\mathbf{A} = \begin{bmatrix} \mathbf{a}_1^T \\ \vdots \\ \mathbf{a}_n^T \end{bmatrix}$ where $\mathbf{a}_i \in \mathbb{R}^n$.

- (b) (7 points) Let $x, y, w \in \mathbb{R}$. Compute and simplify $\frac{\partial f(x, y)}{\partial w}$, where $f(x, y) = x^2 - y^2 + 2xy$, $x(w) = w^2 - 3w$, and $y(w) = w^4 - 3w^2 + w$. Simplify your answer as much as possible and provide a final expression only in terms of w .

Hint: Use the chain rule.

- (c) (7 points) Given an orthogonal matrix $\mathbf{U} \in \mathbb{R}^{n \times n}$, and a vector $\mathbf{x} \in \mathbb{R}^n$. Prove that $\|\mathbf{U}\mathbf{x}\|_2^2 = \|\mathbf{x}\|_2^2$.

- (d) (8 points) For $\mathbf{x} \in \mathbb{R}^n$ and a known symmetric matrix $\mathbf{A} \in \mathbb{S}^{n \times n}$, let $f(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x}$ (the quadratic function). Calculate the gradient $\nabla_{\mathbf{x}} f(\mathbf{x})$. Please show how you calculate the gradient.

Hint: Your final result should only contain $\mathbf{A}$, $\mathbf{x}$ and basic vector operations.

Problem 2 (Probability Review, 20 points) Consider a lottery where the player picks 6 unique numbers from the set $\{1, \dots, 51\}$. The winning numbers are determined by drawing, without replacement, 6 balls uniformly at random from a box containing 51 balls numbered 1 through 51. Order does not matter in determining a winning ticket.

- (a) (10 points) What is the probability, in terms of k , that a purchased ticket matches $k \in \{0, 1, 2, \dots, 6\}$ of the winning numbers?

- (b) (10 points) The player decides she is bored with the lottery and wants to make it more interesting. She builds a machine that will check how many of her numbers match the winning numbers. The machine reads the numbers and determines that there are $0 \leq k \leq 6$ matches, but it does not tell the player what k is. If $k = 0$ the machine tells the player, truthfully, she has 0 matches. If $1 \leq k \leq 6$ the machine rolls k dice and sums the result. If the sum is greater than k , the machine tells the player, truthfully, she has k matches. If the sum is equal to k , the machine lies to the player about her number of matches. What is the probability that the machine lies to the player?

Hint: Use the law of total probability.

Problem 3 (Linear Least Squares, 30 points + 10 bonus points) This problem considers a particular case of the linear least squares problem. Given a square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ and a vector $\mathbf{y} \in \mathbb{R}^n$, our goal is to find a vector $\mathbf{x} \in \mathbb{R}^n$ that minimizes $f(\mathbf{x}) = \|\mathbf{A}\mathbf{x} - \mathbf{y}\|_2^2$. When $\mathbf{A}$ is an invertible square matrix, $\mathbf{y}$ is in the range of $\mathbf{A}$ and $\mathbf{x} = \mathbf{A}^{-1}\mathbf{y}$ is the unique solution. This problem considers the case where $\mathbf{A}$ is

symmetric but *not* invertible. We will use the eigendecomposition of A to derive an analytical solution to linear least squares in this case.

Any diagonalizable matrix A has a decomposition of the form $A = Q\Lambda Q^{-1}$, where $Q, \Lambda \in \mathbb{R}^{n \times n}$, Λ is diagonal, and $Aq_i = \lambda_i q_i$ for all $i = 1, \dots, n$. Here, q_i is the i^{th} column of Q and λ_i is the i^{th} diagonal entry of Λ . We call $A = Q\Lambda Q^{-1}$ the *eigendecomposition* of A . If $A \in \mathbb{R}^{n \times n}$ is a symmetric matrix (which guarantees diagonalizability), the *spectral theorem* guarantees the following:

1. Q has orthonormal columns* (also called a unitary matrix), i.e., $QQ^T = I = Q^T Q$.
2. $\lambda_i \in \mathbb{R}$ for all $i = 1, \dots, n$

Consequently, we can write the eigendecomposition of a real-valued symmetric matrix A as

$$A = Q\Lambda Q^T.$$

NOTE: Since we have assumed A is non-invertible, there exists at least one $i \in \{1, \dots, n\}$ for which $\lambda_i = 0$.

*The spectral theorem actually guarantees the eigenvectors are orthogonal. By convention, we take the eigenvectors to have unit norm. This implies we can write the eigenvalue decomposition as specified.

- (a) (15 points) Prove that $\|Ax - y\|_2^2$ can be written $\|\Lambda\tilde{x} - \tilde{y}\|_2^2$, where $\tilde{x}, \tilde{y}$ are some vectors in $\mathbb{R}^n$. Additionally, give the expressions for $\tilde{x}$ and $\tilde{y}$.

HINT: If $U \in \mathbb{R}^{n \times n}$ is a unitary matrix, it is a fact that $\|Uz\|_2^2 = \|z\|_2^2$ for all $z \in \mathbb{R}^n$. If a norm has this property, it is called unitarily invariant.

- (b) (15 points) Using the above result, prove that $x^* = Q\Lambda^\dagger Q^T y$ is a minimizer of $f(x) = \|Ax - y\|_2^2$, where we take $\Lambda_{i,j}^\dagger = \begin{cases} \frac{1}{\Lambda_{i,j}}, & \Lambda_{i,j} \neq 0 \\ 0, & \text{otherwise} \end{cases}$.

HINT: Write out $\|\Lambda\tilde{x} - \tilde{y}\|_2^2$ as a sum of scalars, and split the sum into terms where the eigenvalues are zero and non-zero.

- (c) (**Optional Challenge Problem, 10 bonus points**) Prove a similar result for much more general matrices A (i.e., not symmetric, not square) using the singular value decomposition. The proof follows a similar approach.

Problem 4 (numpy Exercises, 20 points) Throughout this course we will heavily utilize `numpy` in coding problems on homeworks. These questions are meant to help familiarize you with some useful features of `numpy`. You will apply `numpy` to a few isolated problems. That way you are not learning features (and resolving bugs) in the middle of more complex problems.

`numpy` is an extremely efficient library where many underlying operations are often written in C++. Thus it is often much faster (upwards of 100x in many cases) to perform operations on `numpy` matrices instead of individually on elements using python for loops. For all of these problems do not use loops or list comprehension of any kind.

We have provided a starter jupyter notebook file (`q4_starter.ipynb`) on Canvas to help you work through the following problems. **Please make sure to attach a PDF copy of the completed notebook to your write-up and include it in your page assignments.**

NOTE: Often in python, there is a difference in the way operators and built-in functions treat, for example, size $(N,)$ and size $(N, 1)$ `numpy` arrays (as well as other indexed objects). This can often

lead to (sometimes silent) bugs in code. Best practice is to pay close attention to the size of arrays throughout development and use comments to specify the sizes of inputs and outputs to all your functions. Be sure to follow instructions carefully.

- (a) (10 points) Norms of matrices and vectors have many uses in machine learning, e.g., objective functions, stopping conditions, etc. Given a matrix $\mathbf{A} \in \mathbb{R}^{N \times M}$ (size (N, M) numpy array) return a vector $\mathbf{x} \in \mathbb{R}^N$ (size $(N,)$ numpy array) such that x_i (the i^{th} entry of $\mathbf{x}$) is the ℓ_1 -norm of the i^{th} row of $\mathbf{A}$. Your solution should be a single line.
- (b) (10 points) Another potential application is assigning points to groups. In this question we will consider a set of cell towers and a set of home addresses. The goal is to find the closest cell tower for each home address. You are given the set of cell towers in matrix $\mathbf{A}$. $\mathbf{A} \in \mathbb{R}^{M \times D}$ denotes the locations of M towers in a D -dimensional space. You are also given a matrix $\mathbf{B} \in \mathbb{R}^{N \times D}$ which denotes the locations of N home addresses. You must return the vector $\mathbf{x} \in \mathbb{R}^N$ where x_i is the cell tower that home i (i^{th} row of matrix $\mathbf{B}$) should be assigned to. A buggy solution is provided, which you must fix.

For example, using python indexing notation, if home address $B[8, :]$ is closest to the cell tower $A[6, :]$, then $x[8] = 6$. (maximum lines of code 4 - note that line limits do not include the function definition lines; no partial credit will be awarded if more lines are used).

HINT:

$$\|\mathbf{x} - \mathbf{y}\|_2^2 = \|\mathbf{x}\|_2^2 + \|\mathbf{y}\|_2^2 - 2\mathbf{x}^T \mathbf{y}$$